

Software Requirement Specification (SRS)

English to Hindi Language Translator Web Application

1. Introduction

The purpose of this project is to develop a web-based language translation application that translates English text into Hindi using a Natural Language Processing (NLP) model. The system uses the pretrained Helsinki-NLP/opus-mt-en-hi model available through the Hugging Face Transformers library. The application allows users to enter English text and receive Hindi translation instantly through a simple web interface.

1.2 Scope

The system will accept English text input, translate it to Hindi, display translated Hindi text on the webpage, use a pre-trained NLP model, and provide a simple user-friendly web interface. The system will be deployed online using the Render cloud hosting platform.

1.3 Definitions

NLP: Natural Language Processing SRS: Software Requirement Specification API: Application Programming Interface UI: User Interface Flask: Python web framework Transformer Model: Deep learning model used in NLP

1.4 Technologies Used

Technology	Purpose
Python	Backend programming language
Flask	Web framework
Hugging Face Transformers	NLP model access
HTML	Webpage structure
CSS	Styling of interface
Render	Cloud deployment platform

2. Overall Description

The system is a standalone web application where the user interacts through a browser interface. The frontend sends text input to the Flask backend server, which processes translation using the Hugging Face NLP model. The translated Hindi text is then returned and displayed on the webpage.

2.2 Product Functions

1. Accept English text input 2. Send text to backend server 3. Process translation using NLP model 4. Generate Hindi translation 5. Display translated text on webpage

3. System Features

Feature: Text Translation Input: English text entered by user. Process: Text is processed using the Helsinki NLP translation model. Output: Hindi translated text displayed on the webpage.

Functional Requirements

ID	Requirement
FR1	System shall allow users to enter English text
FR2	System shall send text to backend
FR3	System shall translate text using NLP model
FR4	System shall display Hindi translation
FR5	System shall allow multiple translations

4. External Interface Requirements

User Interface components include: Textbox for entering English text, Translate button to trigger translation, Output box to display translated Hindi text.

5. Non-Functional Requirements

Performance: Translation should occur within 2–5 seconds. Security: No personal data stored and input validation should be applied. Usability: Interface should be simple and easy to understand. Reliability: System should handle translation requests without crashes.

6. Deployment

The application will be deployed using Render. Steps: 1. Push project to GitHub 2. Connect repository to Render 3. Configure Python environment 4. Deploy Flask application 5. Access application via public

URL

7. Future Enhancements

Support multiple languages, add voice translation, implement text-to-speech, improve UI design, and maintain translation history.

8. Conclusion

The English to Hindi Translator Web Application provides a simple and efficient solution for language translation using Natural Language Processing. By integrating Flask, Hugging Face Transformers, and the Helsinki NLP translation model, the system enables fast and accurate translation accessible through a web interface.